PSU ARCHITECTURE

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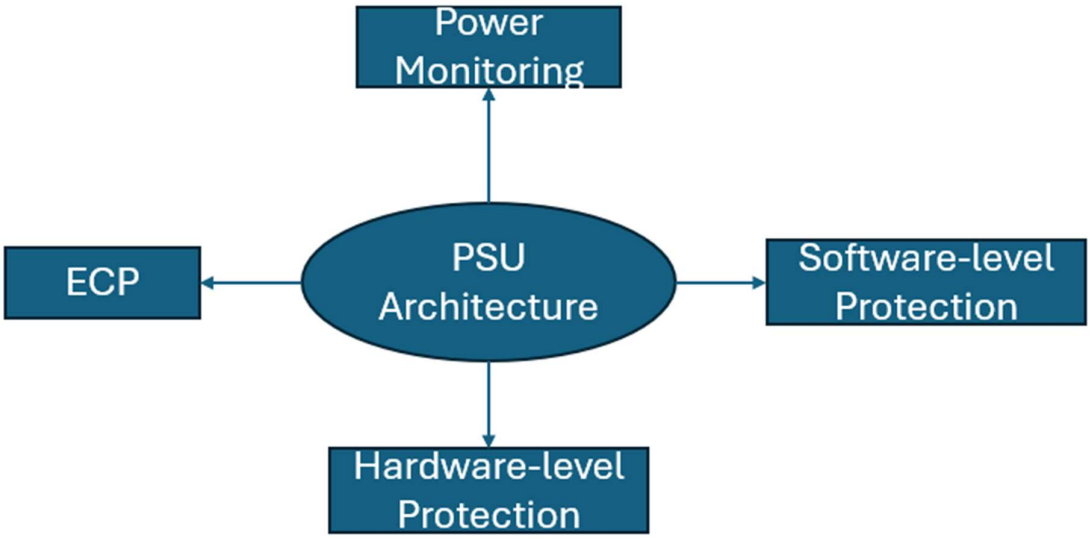
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Introduction

This PSU architecture is designed to make the PSU more efficient and reliable. This PSU architecture was created as the PSU is the main component for electrical safety within a device. This means that if it isn't reliable enough then the device can take damage or stop functioning entirely. This PSU architecture does not change the main processes that current PSUs use. For example, it does not fundamentally change the input stage, PFC, switching stage or output stage. This PSU architecture focuses on communication between the PSU and rest of the system. It also focuses on improving electrical safety of the device.

This PSU architecture cannot promise perfect efficiency or reliability as it is impossible to achieve in any system. However, this PSU architecture promises to try provide a more efficient and reliable PSU compared to today's current models. This PSU architecture was made to be used on most devices that are used today. The technical specifications in this architecture are made to help engineers develop this PSU architecture more easily. Make sure to adapt this PSU architecture to real-world standard and resources.



The spider diagram above shows the 4 main concepts involved within this PSU architecture. ECP stands for electric control protocol which will be used to enforce electrical safety. You can find more information in the sections below.

PSU Logical Electrical Systems

- **These logical systems are designed to allow the motherboard to control the electricity that is provided by the PSU.**
- **There are two subsystems used in this logical electrical control and safety system.**
- **The first subsystem is the electricity control protocol also known as ECP.**
- **ECP is designed to provide the motherboard with full control over the electricity that the PSU provides.**
- **First, the motherboard calculates the voltage that each component needs using current-sensor resistors which I will discuss in the next section.**
- **Second, the motherboard measures the tiny voltage drop within these current-sensor resistors.**
- **This is done by using Ohm's law to calculate the tiny voltage drop and get an instantaneous reading.**
- **Third, the motherboard sums up all the voltages from the calculations and finds the total needed voltage.**
- **Fourth, the motherboard sends a request to the PSU to meet the new voltage requirements.**
- **Fifth, The PSU adapts to the voltage by lowering or increasing its current voltage based on the requirements.**
- **It is important to remember the motherboard does this protocol in real-time using its super I/O chip while the PSU uses its controller for processing.**
- **The second subsystem is the electrical tier system.**
- **This electrical tier system is designed to control the amount of electricity that each type of device/technological infrastructure can have.**
- **There are 4 tiers in total.**
- **For the first tier, the voltage minimal is 50W and the voltage maximum is 200W making it the low power tier.**
- **This tier is designed to power Mini-PCs, networking devices, gaming consoles and small servers.**
- **For the second tier, the voltage minimal is 200W and the voltage maximum is 400W.**
- **This tier is designed for devices like mid-range Gaming PCs and Workstations.**
- **For the third tier, the voltage minimal is 400W and the voltage maximum is 1200W.**
- **This tier is designed for high-end gaming PCs, Workstations and mid-range servers.**
- **For the fourth tier. The voltage minimal is 1200W and the voltage maximum is 2500W.**
- **This tier is designed for high-end servers, data centre racks, high-end workstations and medical imaging devices.**
- **It is important that these can be adjusted to real-world electricity usage.**

PSU Architecture Physical Components and Control Systems

- **These systems are designed to provide the motherboard with the actual components it needs to control the electricity in a worst-case situation.**
- **The first system I will talk about is the physical components.**
- **The first physical component needed is current-sense resistors.**
- **These resistors measure the voltage for each component allowing the motherboard to calculate the voltage drop.**
- **The resistors should be deployed across every power path of electrical components such as RAM, CPU and GPU.**
- **The second physical component is PMBus.**
- **This bus ensures that the PSU can communicate with the motherboard to perform the ECP protocol.**
- **This is important because without communication, the voltage that the PSU provides cannot be adapted.**
- **The third physical component is current limiting circuits.**
- **The current limiting circuits control how much voltage the PSU can provide while giving the motherboard the power to adjust the voltage with hardware if needed.**
- **These circuits should be deployed specifically with the PSU to prevent overvoltage and enforce motherboard requests.**
- **These 3 components were chosen as they provide electricity enforcement, measurement and the communication that's needed for the logical systems.**
- **The second system is user control over electrical usage.**
- **This system is designed to provide the users with some control over how much electricity their device can use.**
- **The user can control what tier their device uses from the BIOS/UEFI interface which will provide the option.**
- **The operating system will calculate what tiers the PSU can support and what tiers that would be safe for the device.**
- **This allows the user to choose what tier they want while also ensuring that it is safe for them to use.**
- **The third system is fallback systems.**
- **This system is designed for two situations that may occur within normal operations of a device.**
- **The first situation is overvoltage especially when the motherboard's request is ignored or failed to reach the PSU.**
- **In this case, the motherboard can use the current limiting circuits to lower the amount of voltage that the PSU can deliver.**
- **The second situation is damage to the components due to overvoltage.**
- **In this case, the motherboard can use the current limiting circuits to completely cut off the electricity delivery that the PSU provides.**

PSU Architecture Workflow

- **This workflow is designed to show the PSU architecture in action from booting, during normal operation, overvoltage and damage to the components due to electrical usage.**
- **This workflow will not be perfect and will differ depending on the device and its configurations.**
- **First, AC input begins for the PSU directly from the wall outlet allowing the PSU to start electricity control processes.**
- **Second, the standby voltage which is 5V is used to boot up the device and provide power until AC input is ready.**
- **Third, the PSU controller is woken up and begins its operations in controlling and delivering electricity to the device.**
- **Fourth, the electricity conversion and transformation process is completed allowing electricity to reach every component easily.**
- **Fifth, the PMBus begins communication with the motherboard giving it data on how much voltage the PSU is processing.**
- **Sixth, the super I/O on the motherboard receives the signal from the motherboard and begins its electricity protocols.**
- **Seventh, the super I/O measures the voltage by using the current sense resistors and Ohm's law.**
- **Eighth, the super I/O sends a request to the PSU asking it to deliver a new voltage of electricity.**
- **Ninith, the PSU adapts to the voltage by increasing the amount of electricity it is delivering.**
- **Tenth, a few hours later, the super I/O uses the ECP protocol once again however the signal fails to reach the PSU.**
- **Eveleth, the PSU continues to deliver an uncontrollable voltage that the components can't handle.**
- **Twelfth, the motherboard uses current limiting circuits to enforce the new voltage limit until communication is reestablished.**
- **Thirteenth, communication is reestablished and the PSU delivers electricity at the requested voltage.**
- **Fourteenth, a few months/years later, the GPU gets damaged slightly from overvoltage.**
- **Fifteenth, the super I/O detects the overvoltage and the damage that the GPU could have taken and acts.**
- **Sixteenth, the super I/O uses the current limiting circuits to limit the voltage and gives the signal to the PSU to begin a shutdown process.**
- **Seventeenth, the PSU is successfully shut down and when the user boots their device, they are told that the GPU may need repairs.**
- **This workflow establishes the resilient and reliable architecture of the PSU as without it the device cannot operate at all.**

PSU Architecture Trade-offs

- **This PSU architecture has trade-offs just like any other system I have created.**
- **Engineers and developers may also face further implementation challenges as each system is different.**
- **I have identified 5 main trade-offs.**
- **The first trade-off is time-consumption.**
- **This trade-off exists because it will take years to implement these hardware changes to the PSU.**
- **This means that the PSU will not be a short-term project and will require consistent resources and effort.**
- **This trade-off can be managed by using structured development processes, working in teams and using existing systems and methods.**
- **The second trade-off is cost.**
- **This trade-off exists because it will mostly likely cost a lot to implement the hardware changes.**
- **This is because implementing resistors and circuits cost physical materials and resources to create.**
- **This trade-off can be managed by using resistors and circuits only where needed, allocating a budget for this project and accepting that this will have its costs.**
- **The third trade-off is coordination.**
- **This trade-off exists because this PSU architecture will require hardware engineers, electrical engineers and OS developers to work together.**
- **This means that there must be a long-term commitment from multiple parties to create this PSU.**
- **This trade-off can be managed by using existing business agreements, clear roles and responsibilities and central oversight to prevent conflicts.**
- **The fourth trade-off is complexity.**
- **This trade-off exists because this PSU architecture can be difficult and confusing to understand and implement.**
- **This is because it requires changes to multiple components like firmware, the motherboard and the PSU itself.**
- **This trade-off can be managed by understanding the architecture, using structured project management processes and consistently testing concepts.**
- **The fifth trade-off is performance.**
- **This trade-off exists because this PSU architecture has limits it can not push all the way to its max form.**
- **This means that power users will not be able to push performance to his limits to the point that it can cause damage.**
- **This trade-off can be managed by allowing users to control voltage within safe limitations and warning them that they may will not be able to push it beyond limits.**